WORKED SOLUTION

STEP III 2024, Question 6

Pure • Linear Differential Equations

STEP III 2024, Question 6

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

The control u is added equally to every equation, so any combination of the coordinates whose coefficients add up to zero doesn't feel u at all. If we choose that combination well, its derivative is a multiple of itself, so it grows or decays exponentially whatever u does. An exponential is never zero unless it starts at zero, which gives the necessary conditions. Once those hold, the motion collapses to a single first-order equation that we can solve with a constant u.

Solution

Part (i)

(a) Subtracting the two equations, the u terms cancel:

$$\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{\mathrm{d}y}{\mathrm{d}t} = (-x + 3y + u) - (x + y + u) = -2(x - y).$$

So $w = x - y$ satisfies $\frac{\mathrm{d}w}{\mathrm{d}t} = -2w$, giving

$$x - y = (x_0 - y_0)\,\mathrm{e}^{-2t}$$

for every function u. If the particle is at the origin at some $t > 0$, then $x - y = 0$ there. Since $\mathrm{e}^{-2t} \neq 0$, this forces $x_0 - y_0 = 0$, that is, $x_0 = y_0$.

(b) With $x_0 = y_0$, the formula in (a) gives $x - y = 0$ for *all* t, so the particle stays on the line $y = x$. Putting $y = x$ into the first equation,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 2x + u.$$

For constant u this is linear with constant coefficients. The particular solution is the constant $-\frac{1}{2}u$, so

$$x = -\tfrac{1}{2}u + \left(x_0 + \tfrac{1}{2}u\right)\mathrm{e}^{2t}.$$

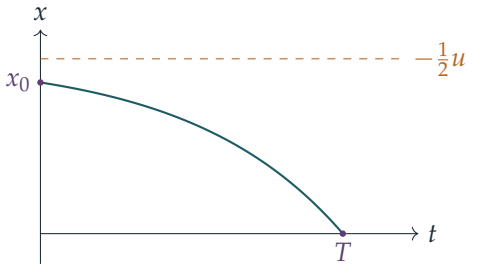
We need $x(T) = 0$ (and then $y(T) = x(T) = 0$ too):

$$\left(x_0 + \tfrac{1}{2}u\right) \mathrm{e}^{2T} = \tfrac{1}{2}u \quad \Longrightarrow \quad x_0 \,\mathrm{e}^{2T} = \tfrac{1}{2}u \left(1 - \mathrm{e}^{2T}\right).$$

Since $T > 0$, $\mathrm{e}^{2T} \neq 1$, so we can divide:

$$\boxed{u = -\frac{2x_0 \,\mathrm{e}^{2T}}{\mathrm{e}^{2T} - 1}}$$

The sign makes sense. Left alone, $\dot{x} = 2x$ pushes the particle away from the origin, so u must push against x_0. (If $x_0 = 0$ the particle is already at rest at the origin with $u = 0$.)



The sketch shows the case $x_0 = 1$, $T = 1$. The particle starts just below the unstable equilibrium $x = -\frac{1}{2}u$ and is driven away from it, reaching the origin exactly at time T.

Part (ii)

(a) Following part (i), we look for $ax + by + cz$ with $a + b + c = 0$ (so u cancels) whose derivative is a multiple of itself. The pattern "x, then $-2y$, then z" suggests $w = x - 2y + z$:

$$\begin{aligned}\frac{\mathrm{d}w}{\mathrm{d}t} &= (4y - 5z + u) - 2(x - 2z + u) + (x - 2y + u) \\ &= -x + 2y - z = -w.\end{aligned}$$

Hence $x - 2y + z = (x_0 - 2y_0 + z_0)\,\mathrm{e}^{-t}$, whatever u is. At the origin $w = 0$, and $\mathrm{e}^{-t} \neq 0$, so $x_0 - 2y_0 + z_0 = 0$. That is, $y_0 = \frac{1}{2}(x_0 + z_0)$, the mean of x_0 and z_0.

? **Ask yourself** Is there a second combination that also ignores u?

The last two equations both contain $x + u$, so subtracting them removes x as well as u. Let $v = y - z$. Then

$$\frac{\mathrm{d}v}{\mathrm{d}t} = (x - 2z + u) - (x - 2y + u) = 2(y - z) = 2v,$$

so $y - z = (y_0 - z_0)\,\mathrm{e}^{2t}$.

(b) If the particle is at the origin at some $t > 0$, then $v = 0$ there. Since $\mathrm{e}^{2t} \neq 0$, we get $y_0 = z_0$. Combining this with (a), $x_0 = 2y_0 - z_0 = y_0$, so $x_0 = y_0 = z_0$.

(c) With $x_0 = y_0 = z_0$, both w and v start at 0, so they stay 0 for all t: $y = z$, and then $x = 2y - z = y$. The particle moves along the line $x = y = z$, and each equation becomes

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 4x - 5x + u = -x + u$$

(the second and third give $x - 2x + u$, the same). For constant u,

$$x = u + (x_0 - u)\,\mathrm{e}^{-t}.$$

Setting $x(T) = 0$ gives $u\left(1 - \mathrm{e}^{-T}\right) = -x_0\,\mathrm{e}^{-T}$. Since $T > 0$ we may divide, and multiplying top and bottom by e^{T},

$$\boxed{u = -\frac{x_0}{\mathrm{e}^{T} - 1}}$$

Here the uncontrolled motion $\dot{x} = -x$ already decays towards the origin, but only as $t \to \infty$. A small push towards the origin, which tends to zero as $T \to \infty$, gets the particle there in finite time.

Checks

- Limits: as $T \to \infty$ the answer to (ii)(c) tends to 0, matching the natural decay, while the answer to (i)(b) tends to $-2x_0$, which makes $x_0 = -\frac{1}{2}u$ an equilibrium. The particle then sits there and never arrives, which is why a finite T needs u slightly larger in size. As $T \to 0^{+}$ both answers blow up, as they should.
- Numerical: with $x_0 = 1$, $T = 1$, part (i)(b) gives $u \approx -2.3130$, and stepping the equations forward numerically lands on $(0, 0)$ at $t = 1$ to within 10^{-8}. The same holds for (ii)(c) with $u = -1/(\mathrm{e} - 1) \approx -0.5820$. The invariants $x - y$, $x - 2y + z$ and $y - z$ were also confirmed to follow their exponentials with u a non-constant function of time.

Common slips

- Assuming u is constant in (i)(a) or (ii)(a),(b). Those parts are for *any* u, and the cancellation of u is the whole point.
- In (i)(b) and (ii)(c), forgetting to say *why* $y = x$ (or $x = y = z$) for all time. It follows from the exponential formulas with zero initial value.

Beyond STEP

This is a first taste of *control theory*. Writing the system as $\dot{\mathbf{r}} = A\mathbf{r} + \mathbf{b}u$, the combinations we found are left eigenvectors of A that are orthogonal to $\mathbf{b}$, so no control can affect them. Kalman's rank condition decides whether every state can be steered to the origin. Here it fails, which is exactly why only special starting points work. Strang's

Introduction to Linear Algebra covers the eigenvector side, and Boyce and DiPrima's *Elementary Differential Equations* covers linear systems of ODEs.

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